

TASK 2 – GATE-LEVEL SYNTHESIS & SIMULATION

8-Bit ALU using Yosys and Icarus Verilog

Digital VLSI Internship Assignment Report

Project	8-Bit ALU Gate-Level Synthesis
RTL	alu_8bit.sv
Synthesis Tool	Yosys
Simulation Tool	Icarus Verilog
Generated Netlist	alu_gate.v
Total Tests	110
PASS	110
FAIL	0

1. Objective

The objective of this task is to synthesize the verified 8-bit ALU RTL into a gate-level Verilog netlist and verify the generated netlist using gate-level functional simulation. This demonstrates the transition from RTL description to synthesized logic.

2. Synthesis Flow

Yosys was used as the open-source synthesis tool. The SystemVerilog ALU source was read, the design hierarchy was established with `alu_8bit` as the top module, RTL processes were converted, optimization passes were applied, technology mapping was performed, and the synthesized design was written to a Verilog netlist.

The synthesis script used the following main commands: `read_verilog -sv, hierarchy -top alu_8bit, proc, opt, memory, techmap`, and `write_verilog -noattr alu_gate.v`.

3. Generated Gate-Level Netlist

The synthesis process generated `alu_gate.v`. This file represents the ALU using synthesized gate-level structures rather than the original high-level SystemVerilog RTL. The generated netlist was then used as the design under test for gate-level simulation.

4. Gate-Level Functional Simulation

The functional testbench was used to exercise the synthesized netlist. Directed tests and random tests were applied, and the outputs were compared against the expected reference-model values. This provides a functional comparison between the original RTL behavior and the synthesized implementation.

5. Verification Results

Metric	Result
Total tests executed	110
Passed tests	110
Failed tests	0
ALU operations covered	All 8
Overall status	ALL TESTS PASSED

6. Synthesis and Simulation Evidence

Figure 1 – Yosys synthesis console / successful synthesis

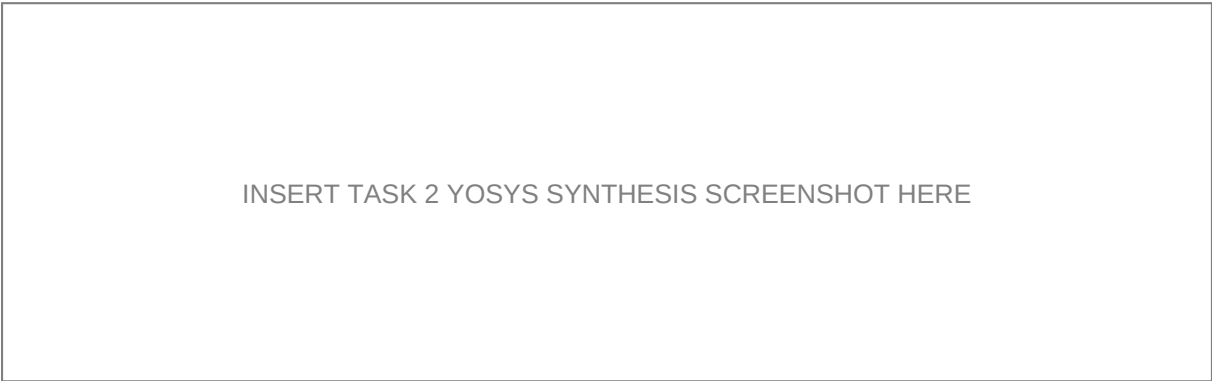


Figure 2 – Gate-level simulation result

INSERT TASK 2 GATE-LEVEL SIMULATION SCREENSHOT HERE

7. RTL vs Gate-Level Comparison

The gate-level simulation produced the same functional verification outcome as the RTL-level simulation: 110 tests passed and zero tests failed. All eight ALU operations were exercised. This provides functional evidence that the synthesized netlist preserves the tested behavior of the original RTL implementation.

8. Deliverables

- **alu_8bit.sv** – synthesizable ALU RTL
- **synth.js** – Yosys synthesis script
- **alu_gate.v** – synthesized gate-level netlist
- **alu_8bit_tb.sv** – functional testbench
- Yosys synthesis evidence
- Gate-level simulation evidence
- This report

9. Conclusion

The 8-bit ALU was successfully synthesized using Yosys and the generated gate-level netlist was functionally simulated using Icarus Verilog. The verification completed 110 tests with 110 passes and zero failures, covering all eight ALU operations. The task demonstrates a basic RTL-to-gate-level synthesis and verification workflow.

Submission note: Replace the two screenshot placeholders with the actual Task 2 screenshots before final submission if the evaluator requires screenshots embedded inside the PDF.